

(accessible everywhere) and local (accessible only within a block / function).

Compilers represent scope using :-

i) Nested Symbol Tables :-

- Each block has its own symbol table.
- Tables are arranged in a hierarchical (tree-like) structure.
- Each table has a pointer to its parent (outer scope).

ii) Stack of Symbol Tables :-

- When entering a block $\rightarrow$ push new table.
- When exiting $\rightarrow$ pop table.
- Always search from top (current scope) to bottom.

Nested Blocks Example :-

{

int x=10;

{

int y=20; // inner scope

int x=30; // shadows outer x

}

}

Outer Scope Table :-

x $\rightarrow$ int (value=10)

Inner Scope Table :-

y $\rightarrow$ int (value=20)

x $\rightarrow$ int (value=30) // shadows outer x

- Inside inner block: x=30 is used (local).